

SHACHE, China

WMO: 518110

Lat: 38.433N

Lon: 77.267E

Elev: 1232

StdP: 87.37

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-11.8	-19.2	0.8	-7.7	-16.9	1.0	-6.2	3.5	0.3	3.3	-0.3	0.6	270	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.6	18.5	33.2	18.0	32.0	17.7	21.8	29.5	20.6	28.9	19.7	28.4	1.9	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	16.5	26.3	18.0	15.0	25.1	16.8	13.9	24.4	70.5	29.5	65.7	29.0	62.2	28.3	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.1	4.1	3.4	-15.1	37.3	3.3	1.1	-17.5	38.1	-19.5	38.8	-21.4	39.4	-23.8	40.2		
			-15.6	23.6	3.3	2.1	-17.9	25.1	-19.8	26.3	-21.7	27.5	-24.0	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-5.0	0.7	9.6	16.5	20.6	24.3	25.6	24.3	19.9	13.2	5.2	-2.6
	DBStd	10.98	3.60	3.81	4.10	3.67	3.09	2.62	2.50	2.39	2.51	3.26	3.62	3.00
	HDD10.0	1334	465	260	59	4	0	0	0	0	0	8	148	390
	HDD18.3	2793	723	493	271	74	15	1	0	1	13	161	394	648
	CDD10.0	2338	0	0	47	199	329	428	484	442	296	108	4	0
	CDD18.3	757	0	0	1	19	86	179	227	184	59	2	0	0
	CDH23.3	7685	0	0	9	259	844	1856	2407	1724	543	43	0	0
	CDH26.7	3081	0	0	1	63	266	796	1113	702	137	2	0	0
Wind	WSAvg	1.4	1.0	1.2	1.6	1.8	1.8	1.8	1.6	1.4	1.3	1.0	0.9	0.9
Precipitation	PrecAvg	47	1	2	2	4	9	8	6	7	3	2	2	1
	PrecMax	125	6	13	27	44	42	41	22	61	17	30	20	6
	PrecMin	11	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	31	2	3	5	10	12	11	6	13	5	6	4	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	13.7	24.5	31.0	32.6	35.5	36.9	35.5	31.5	26.6	20.1	10.4
		MCWB	1.2	4.3	9.8	13.8	15.5	17.9	19.7	20.3	17.3	13.3	8.2	2.5
	2%	DB	4.7	11.3	21.7	28.5	31.1	33.9	35.2	33.3	29.8	24.5	16.9	6.2
		MCWB	-1.0	3.3	8.5	12.8	14.9	16.9	19.0	19.1	16.4	12.8	6.8	0.7
	5%	DB	2.5	9.3	19.6	26.4	29.7	32.5	33.6	31.9	28.2	22.6	14.5	3.9
		MCWB	-2.1	2.3	7.8	12.0	14.7	16.6	18.8	19.2	16.0	12.0	5.9	-0.7
	10%	DB	0.5	7.2	17.5	24.3	28.0	31.1	32.2	30.6	26.6	20.6	12.1	2.1
		MCWB	-3.3	1.1	6.9	11.0	14.1	16.4	18.4	18.8	15.5	10.7	5.0	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	5.7	10.3	14.7	18.5	20.7	23.7	23.6	20.3	15.1	9.0	3.0
		MCDB	8.4	12.0	23.0	28.1	28.0	29.5	30.5	31.1	27.6	22.9	18.3	9.6
	2%	WB	-0.8	3.9	9.1	13.6	16.8	19.1	21.9	21.9	18.4	13.7	7.4	0.9
		MCDB	4.3	10.1	20.4	26.3	26.9	28.7	29.6	29.5	26.0	22.3	15.9	5.5
	5%	WB	-2.0	2.6	8.1	12.6	15.7	18.1	20.7	20.7	17.3	12.7	6.2	-0.4
		MCDB	2.1	8.6	18.4	24.4	26.3	28.7	29.3	28.5	24.8	21.0	13.4	3.5
	10%	WB	-3.1	1.4	7.2	11.7	14.9	17.4	19.9	19.7	16.5	11.6	5.2	-1.5
		MCDB	0.2	6.8	16.5	22.7	25.8	28.2	29.0	27.9	23.9	19.0	11.7	1.9
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.1	11.2	12.1	12.2	12.1	11.7	11.3	12.4	14.2	12.5	9.2
		MCDBR	12.6	12.8	14.1	16.1	15.1	14.2	14.0	13.8	15.1	15.7	16.2	11.6
		MCWBR	8.1	6.9	6.1	6.1	5.9	5.2	5.4	5.8	6.1	6.8	8.0	7.1
	5% WB	MCDBR	12.0	12.1	13.4	14.4	13.4	12.6	12.6	12.7	13.6	14.7	15.8	11.2
		MCWBR	7.8	6.7	5.9	5.7	6.0	5.4	6.0	6.4	6.5	6.9	7.9	6.9
Clear-Sky Solar Irradiance	taub	0.355	0.417	0.546	0.607	0.618	0.610	0.601	0.652	0.632	0.520	0.432	0.381	
	taud	2.060	1.924	1.642	1.551	1.584	1.632	1.674	1.578	1.573	1.727	1.878	1.986	
	Ebn at Noon	785	777	706	696	701	707	709	656	629	658	677	711	
	Edh at Noon	120	158	233	271	267	254	242	260	247	191	144	121	
All-Sky Solar Radiation	RadAvg	2.60	3.48	4.80	5.78	6.70	7.12	6.76	5.97	5.20	4.32	3.12	2.43	
	RadStd	0.38	0.33	0.30	0.20	0.37	0.29	0.37	0.39	0.30	0.16	0.15	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.45	N/A	N/A	+0.52	-1.38	-2.08	N/A	-112	+134	+69
Regional (3 neighbors)		+0.32	N/A	N/A	N/A	-1.04	-1.26	N/A	N/A	+121	+54

Nomenclature: See separate page